Derivation and Implementation of a Multi-Mass Spring Patch Coupled to an Inverted-Flag

Amber Parker and Mike Yan
University of Illinois Urbana-Champaign

This work investigates the coupling of an Euler-Bernoulli beam with a spring-mass patch consisting of an internal chain of masses and springs attached to the beam. This system is designed for integration into an inverted-flag fluid-structure interaction (FSI) solver. The project involves deriving the governing equations of motion (EOMs), developing numerical implementations in Julia, and validating the model.

Nomenclature

$w \in \mathbb{R}^m$	Beam displacement vector (global degrees of freedom)
$\dot{w}, \ddot{w}$	Beam velocity and acceleration
w_j	Beam displacement at node j
$v^{(j)} \in \mathbb{R}^{n_j}$	Relative displacement vector of chain attached at node j
$\dot{v}^{(j)}, \ddot{v}^{(j)}$	Velocity and acceleration of chain j
u_i	Absolute displacement of the i-th mass in a chain
v_i	Relative displacement of the i-th mass with respect to the beam
$M_b \in \mathbb{R}^{m\times m}$	Beam mass matrix
$K_b \in \mathbb{R}^{m\times m}$	Beam stiffness matrix
$M_c^{(j)} \in \mathbb{R}^{n_j\times n_j}$	Mass matrix of chain j
$K_c^{(j)} \in \mathbb{R}^{n_j\times n_j}$	Stiffness matrix of chain j
$m_i^{(j)}$	Mass of the i-th element in chain j
$k_i^{(j)}$	Spring stiffness between masses in chain j
$k_0^{(j)}, k_{n_j}^{(j)}$	Endpoint spring stiffnesses of chain j
$\mathbf{b}_j \in \mathbb{R}^m$	Selection vector mapping forces to beam node j
$\mathbf{1}_{n_j}$	Vector of ones of size n_j

$F_{\text{ext}} \in \mathbb{R}^m$	External force vector acting on the beam
$F_{\text{chain}\to\text{beam}}$	Reaction force from a chain applied to the beam
$S \subseteq \{1, \ldots, m\}$	Set of beam nodes with attached chains
n_j	Number of masses in chain j
m	Number of beam degrees of freedom

I. Introduction

This work summarizes the development of a mathematically consistent and numerically robust model for coupling a multi-mass spring chain to an Euler–Bernoulli (EB) beam. In this context, the introduction of a spring-mass "patch" provides a framework for capturing localized inertia and stiffness effects that are not represented in classical beam formulations. These internal dynamics can significantly alter the global response of the structure, including its stability, frequency content, and energy transfer characteristics.

This work derives the governing equations of motion for a beam coupled to one or more multi-degree-of-freedom spring-mass chains. The derivation is carried out by decomposing the motion of each chain into a beam-driven rigid component and a relative internal deformation, which leads to a clear separation of beam and patch dynamics. The resulting formulation reveals that coupling occurs through both inertial forcing and boundary reaction forces at the attachment points.

The full implementation developed for this work is available in the accompanying GitHub repository: `https://github.com/amberkp/Research`.

II. Derivation

A. One Spring-Mass Chain at One Beam Node

We begin by considering a single spring-mass chain attached to the beam at node j.

For $i = 1, 2, ..., n$, let the displacement of the i-th mass in the chain be

$$u_i = w_j^* + v_i, \tag{1}$$

where w_j^* is the displacement of the beam at node j, and v_i represents the relative displacement of the mass with respect to the beam. This decomposition separates the global beam motion from the internal dynamics of the chain.

Taking differences between adjacent masses eliminates the beam displacement:

$$u_{i+1} - u_i = (w_j^* + v_{i+1}) - (w_j^* + v_i) \tag{2}$$

$$= v_{i+1} - v_i. \tag{3}$$

Thus, the internal spring forces depend only on the relative coordinates v_i.

Following Newton's second law, the equations of motion for each mass are given by:

$$m_i \ddot{u}_i = k_{i-1}(u_{i-1} - u_i) + k_i(u_{i+1} - u_i). \tag{4}$$

Substituting $u_i = w_j^* + v_i$ yields

$$m_i \ddot{v}_i = k_{i-1}(v_{i-1} - v_i) + k_i(v_{i+1} - v_i) - m_i \ddot{w}_j^*. \tag{5}$$

The term $-m_i \ddot{w}_j^*$ represents an inertial forcing due to the acceleration of the beam, which drives the internal dynamics of the chain.

For the endpoints, $u_0 = u_{n+1} = w_j^*$, meaning both ends of the chain are attached directly to the beam. This implies Dirichlet boundary conditions on the relative displacement:

$$v_0 = v_{n+1} = 0.$$

The chain dynamics can be written in matrix form as

$$\begin{bmatrix} m_1 & 0 & \cdots & 0 \\ 0 & m_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_n \end{bmatrix} \ddot{\mathbf{v}} + \begin{bmatrix} k_0 + k_1 & -k_1 & 0 & \cdots & 0 \\ -k_1 & k_1 + k_2 & -k_2 & \cdots & 0 \\ 0 & -k_2 & k_2 + k_3 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -k_{n-1} \\ 0 & 0 & \cdots & -k_{n-1} & k_{n-1} + k_n \end{bmatrix} \mathbf{v} = - \begin{bmatrix} m_1 & 0 & \cdots & 0 \\ 0 & m_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_n \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \ddot{w}_j^* \tag{6}$$

or

$$M_c \ddot{v} + K_c v = -M_c \mathbf{1}\, \ddot{w}_j^*, \tag{7}$$

Thus, the chain behaves as a forced linear system driven by the beam acceleration.

Only the endpoint springs transmit force back to the beam:

$$f_0 = k_0(u_1 - w_j^*) = k_0 v_1, \tag{8}$$

$$f_n = k_n(u_n - w_j^*) = k_n v_n. \tag{9}$$

Thus, the total force from the chain on the beam is

$$F_{\text{chain}\to\text{beam}} = k_0 v_1 + k_n v_n. \tag{10}$$

We can now incorporate this into the discretized beam equation:

$$M_b \ddot{\mathbf{w}} + K_b \mathbf{w} = \mathbf{F}_{ext} + \mathbf{b}(k_0 v_1 + k_n v_n), \tag{11}$$

where $\mathbf{b} \in \mathbb{R}^m$ is a selection vector with a 1 at the jth entry and zeros elsewhere. This vector maps the scalar chain force into the appropriate beam degree of freedom.

B. Deriving Generalized EOMs

Let $S \subseteq \{1, ..., m\}$ be the set of beam nodes that carry can carry a mass-spring chain, which excludes fixed supports. Hence, each node $j \in S$ has its own chain with varying sizes. That is, each chain j has its own mass matrix denoted by $M_c^{(j)} = \mathbf{diag}\left(m_1^{(j)}, ..., m_{n_j}^{(j)}\right)$ with stiffness matrix $K_c^{(j)}$ of size $n_j \times n_j$. So each chain must come with its own respective selection vector, $\mathbf{b}_j$, meaning $\mathbf{b}_j^T \mathbf{w} = w_j$.

We can borrow (7) to yield each chain at node j's EOM:

$$M_c^{(j)} \ddot{\mathbf{v}}^{(j)} + K_c^{(j)} \mathbf{v}^{(j)} = -M_c^{(j)} \mathbf{1}_{n_j} \ddot{w}_j \tag{12}$$

where $\ddot{w}_j = \mathbf{b}_j^T \ddot{\mathbf{w}}$. We can see from this equation that each chain is independent to any other chain, and can only interact through the beam deflection node to which they are attached.

Similarly, we see that our beam EOM will now include multiple reaction forces summed with respect to $j \in S$:

$$M_b \ddot{\mathbf{w}} + K_b \mathbf{w} = \mathbf{F}_{ext} + \sum_{j \in S} \mathbf{b}_j (k_0^{(j)} v_1^{(j)} + k_{n_j}^{(j)} v_{n_j}^{(j)}) \tag{13}$$

1. Equivalent Form of Beam EOM with Inertial Forcing

Equivalently, we can construct the inertial form from this reaction force perspective. We will later see that this will produce a diagonal global stiffness matrix with off-diagonal coupling blocks in the global mass matrix as opposed to the

vice-versa. The inertial form can be constructed by summing the forces (4) within a spring-mass chain in the absolute frame of reference:

$$\sum_{i=1}^{n} m_i \ddot{u}_i = k_0(u_0 - u_1) + k_n(u_{n+1} - u_n) = -(k_0 v_1 - k_n v_n) \tag{14}$$

since this series is telescopic, that is, all internal spring forces appear twice with opposing signs, leaving only the end forces. Additionally we know that (14) represents the force the beam exerts on the chain, supporting our net force on the beam equation (10). Now, we can rewrite the RHS of (14) as

$$\sum_{i=1}^{n} m_i \ddot{u}_i = M_c^{tot} \ddot{w}_j - \mathbf{1}^T M_C \ddot{\mathbf{v}} \tag{15}$$

where $M_c^{tot} = \sum_{i=1}^{n} m_i$. This can simply be extended to our generalized configuration $j \in S$, which gives the equivalent form of (13), our beam EOM, as

$$\left(M_b + \sum_{j \in S} M_{c,j}^{tot} \mathbf{b}_j \mathbf{b}_j^T \right) \ddot{\mathbf{w}} + \sum_{j \in S} \mathbf{b}_j \mathbf{1}_{n_j}^T M_c^{(j)} \ddot{\mathbf{v}}^{(j)} + K_b \mathbf{w} = \mathbf{F_{ext}} \tag{16}$$

2. Global System Assembly

Now we can define our global state vector as

$$\mathbf{q} = \begin{bmatrix} \mathbf{w} \\ \mathbf{v}^{(j_1)} \\ \mathbf{v}^{(j_2)} \\ \vdots \end{bmatrix} \qquad \mathcal{M} = \begin{bmatrix} M_b + \sum_{j \in \mathcal{S}} M_{c,j}^{\text{tot}} \mathbf{b}_j \mathbf{b}_j^\top & \mathbf{b}_{j_1} \mathbf{1}^\top M_c^{(j_1)} & \mathbf{b}_{j_2} \mathbf{1}^\top M_c^{(j_2)} & \cdots \\ M_c^{(j_1)} \mathbf{1}\, \mathbf{b}_{j_1}^\top & M_c^{(j_1)} & \mathbf{0} & \cdots \\ M_c^{(j_2)} \mathbf{1}\, \mathbf{b}_{j_2}^\top & \mathbf{0} & M_c^{(j_2)} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{17}$$

$$\mathcal{K} = \begin{bmatrix} K_b & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & K_c^{(j_1)} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & K_c^{(j_2)} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{18}$$

$$\mathcal{M} \ddot{\mathbf{q}} + \mathcal{K} \mathbf{q} = \mathcal{F} \tag{19}$$

where $\mathcal{F}$ is the external force vector, not including any chain systems.

III. Implementation and Validation

The governing equations are implemented in `Julia`, and the assembled system is integrated using `OrdinaryDiffEq`. Boundary conditions are enforced by removing constrained degrees of freedom, and natural frequencies and mode shapes are obtained from the generalized eigenvalue problem.

To verify the correctness of the implementation, a series of limiting cases are evaluated. Each case is selected such that the expected physical behavior is known *a priori*, allowing the numerical results to be directly compared against well-understood system behavior. For each case, validation is performed using both time-domain response plots and the computed natural frequencies.

A. Euler–Bernoulli Recovery Limit

In this test case, the spring stiffness is set to a very large value while the attached masses are made very small relative to the beam. This configuration is chosen because, in the limit of infinite stiffness and negligible mass, the spring-mass chains should not introduce any meaningful dynamic effect. Physically, the chains become effectively rigid with negligible inertia, and the system should reduce to a standalone Euler–Bernoulli beam.

This case is a valid validation test because the beam-only solution is known and serves as a baseline reference. The beam displacement response and mode shapes obtained from the coupled model are compared directly with the Euler–Bernoulli solution. As shown in the corresponding response plot and modal results, the displacement histories overlap and the natural frequencies match closely. This confirms that the coupling terms do not introduce spurious stiffness or inertia and that the beam formulation is implemented correctly.

B. Decoupled Chain Limit (Soft Springs)

In this case, the spring stiffness is reduced to a very small value. This test is chosen because it represents the opposite extreme of the coupling behavior. Physically, very soft springs transmit negligible force, so the chains should not significantly influence the beam dynamics.

This is a valid validation case because, in this limit, the beam should again behave like a standalone Euler–Bernoulli beam, while the chains move largely independently. The beam response plots confirm that the displacement behavior is nearly identical to the uncoupled beam case. Additionally, the computed natural frequencies match the beam-only frequencies, indicating that the coupling terms correctly vanish as the spring stiffness approaches zero. The chain displacement plots show increased relative motion, which is consistent with the expected weak coupling.

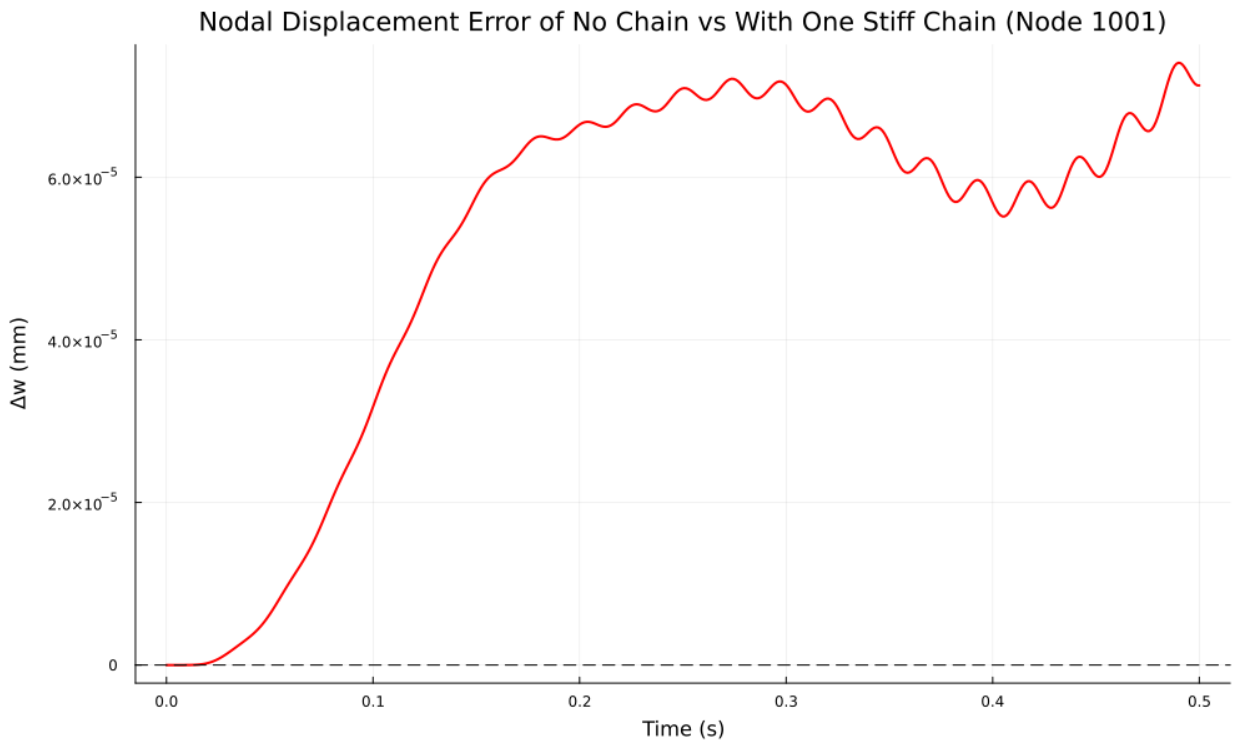

Fig. 1 Plot of the tip displacement difference of Pure Euler Bernoulli Beam vs Euler Bernoulli Beam with one chain located at free end built with two very stiff springs and minuscule mass. A ramped static load is placed at the tip.

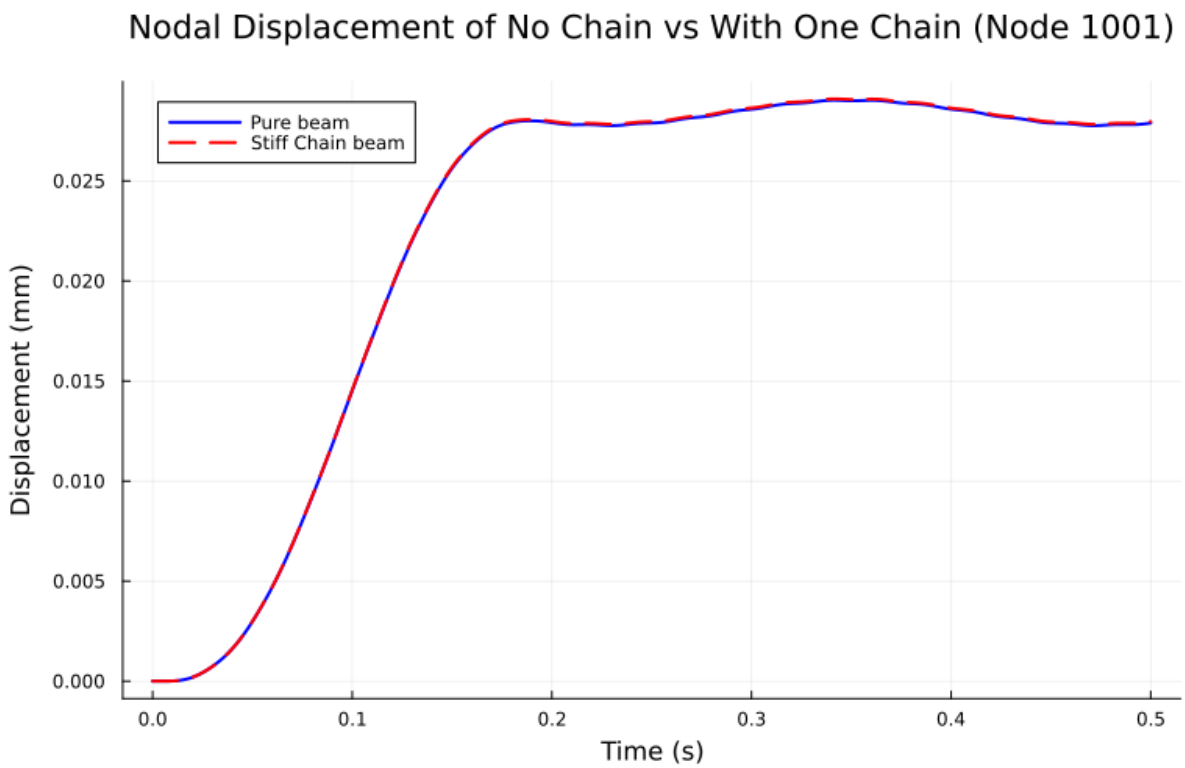

Fig. 2 Plot of the tip displacement of Pure Euler Bernoulli Beam vs Euler Bernoulli Beam with one chain located at free end built with two very stiff springs and minuscule mass. A ramped static load is placed at the tip.

C. Added Mass Limit

In this case, the spring stiffness is kept large while the attached masses are increased. This configuration is selected to test whether the model correctly captures inertial coupling. When the springs are stiff, the chain masses move approximately with the beam at the attachment node, effectively acting as lumped masses attached to the beam.

This case is valid because the expected behavior is well known: increasing the system mass should reduce the natural frequencies. The modal results show a clear decrease in the natural frequencies compared to the baseline beam, consistent with the added inertia. The displacement plots also reflect the increased mass through slower dynamic response. This confirms that the mass coupling between the beam and the chains is implemented correctly.

D. Summary of Validation

Across all test cases, the numerical results match the expected physical behavior in each limiting regime. The Euler–Bernoulli recovery case verifies the beam formulation, the soft-spring case verifies proper decoupling, the added-mass case verifies inertial coupling, and the resonant case verifies dynamic interaction. The agreement between expected and observed trends in both the response plots and modal results provides strong confidence in the correctness of the implementation.

Mode	Pure Beam (Hz)	With Chain (Hz)	Change (%)
1	41.837	16.205	−61.27
2	262.629	193.437	−26.34
3	735.406	606.247	−17.56
4	1441.102	1252.995	−13.05
5	2382.246	2134.148	−10.41
6	3558.665	3249.161	−8.70
7	4970.366	4597.298	−7.51
8	6617.351	6177.593	−6.65
9	8499.619	7988.809	−6.01
10	10617.171	10029.382	−5.54
11	12970.007	12297.326	−5.18
12	15558.126	14790.106	−4.94
13	18381.529	17504.480	−4.77
14	21440.215	20436.321	−4.68
15	24734.185	23580.454	−4.67
16	28263.439	26930.638	−4.71
17	32027.976	30479.921	−4.83
18	36027.797	34221.761	−5.01
19	40262.901	38152.222	−5.24
20	44733.289	42272.875	−5.50

Table 2 First 20 natural frequencies of the pure Euler–Bernoulli beam versus the beam with a 20 kg chain mass at the free tip (stiff springs), showing the reduction due to added inertia.

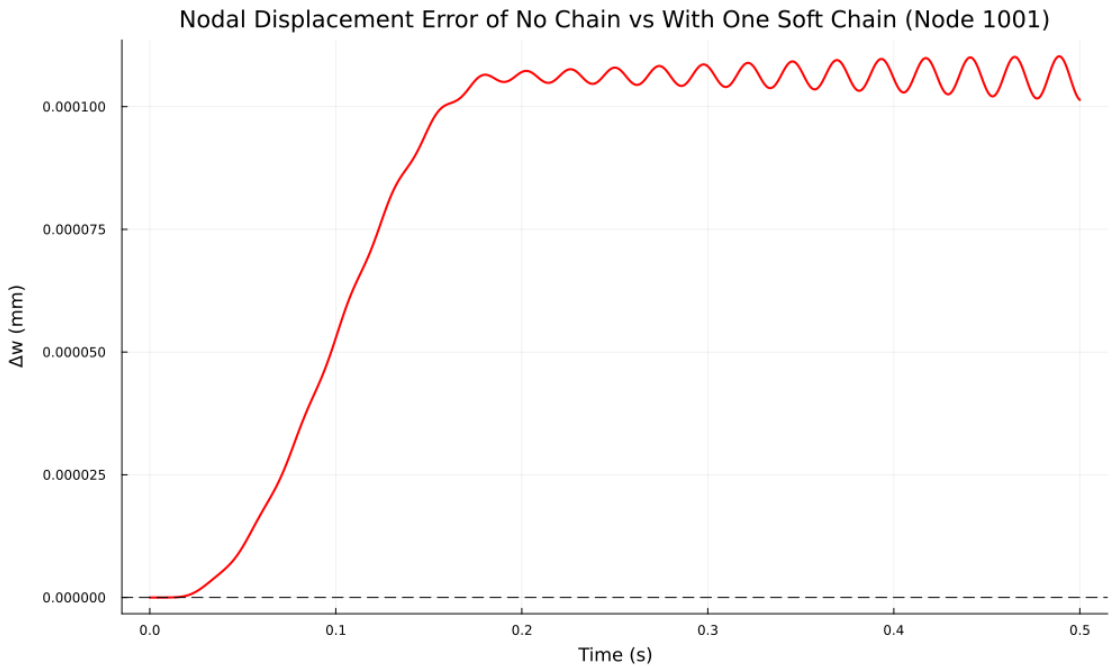

Fig. 3 Plot of the tip displacement difference of Pure Euler Bernoulli Beam vs Euler Bernoulli Beam with one chain located at free end built with two very weak springs and minuscule mass. A ramped static load is placed at the tip.

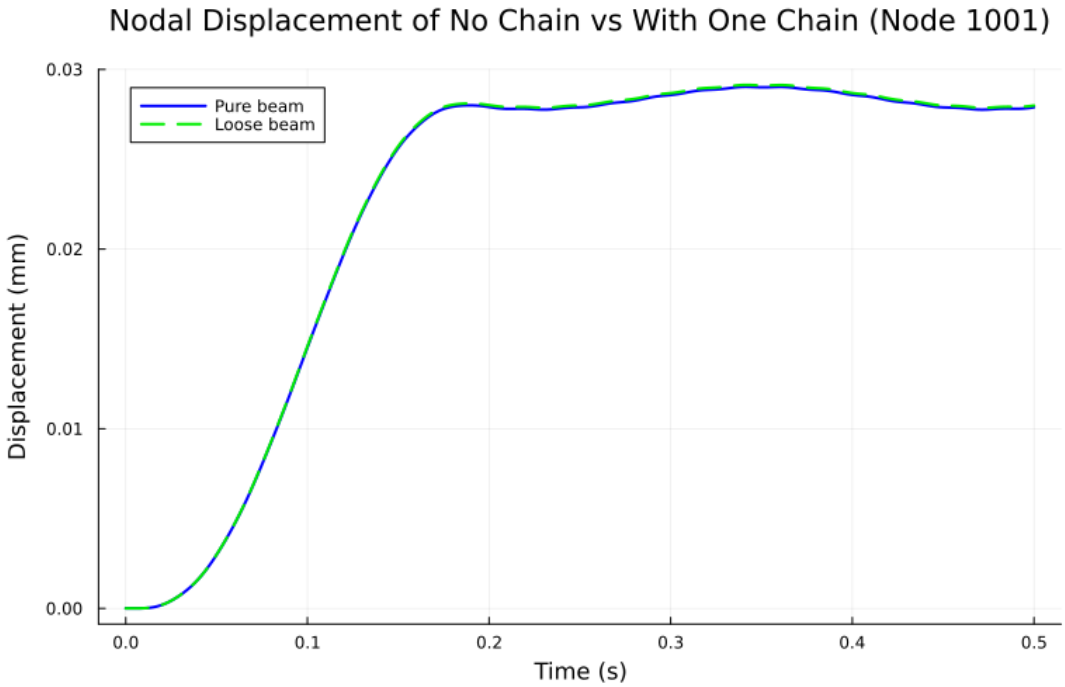

Fig. 4 Plot of the tip displacement of Pure Euler Bernoulli Beam vs Euler Bernoulli Beam with one chain located at free end built with two very weak springs and minuscule mass. A ramped static load is placed at the tip.

IV. Conclusion

This work developed and implemented a generalized framework for coupling discrete multi-mass spring chains to an Euler–Bernoulli beam. A mathematically consistent formulation was derived for both single and multiple attached spring-mass patches, capturing the inertial and reaction-force coupling between the beam and the internal chain dynamics. The resulting global system was implemented numerically in Julia, enabling both time-domain simulation and modal analysis of coupled configurations with arbitrary patch placement and parameter selection.

Validation through limiting-case analysis demonstrated that the implementation reproduces expected physical behavior in known regimes, including recovery of the standalone Euler–Bernoulli beam response, decoupling in the weak-coupling limit, and inertial frequency shifts due to added mass effects. These results provide confidence in the correctness and robustness of the implementation.

The developed framework establishes a flexible foundation for future integration into the intended inverted-flag fluid–structure interaction solver, where localized internal dynamics may significantly influence global structural response.

Acknowledgments

The authors would like to acknowledge the use of AI tools, including ChatGPT and Claude, which were used to support code refinement, debugging, and LaTeX formatting during the preparation of this work. All technical derivations, implementation decisions, and validation methodology were reviewed and verified by the authors.